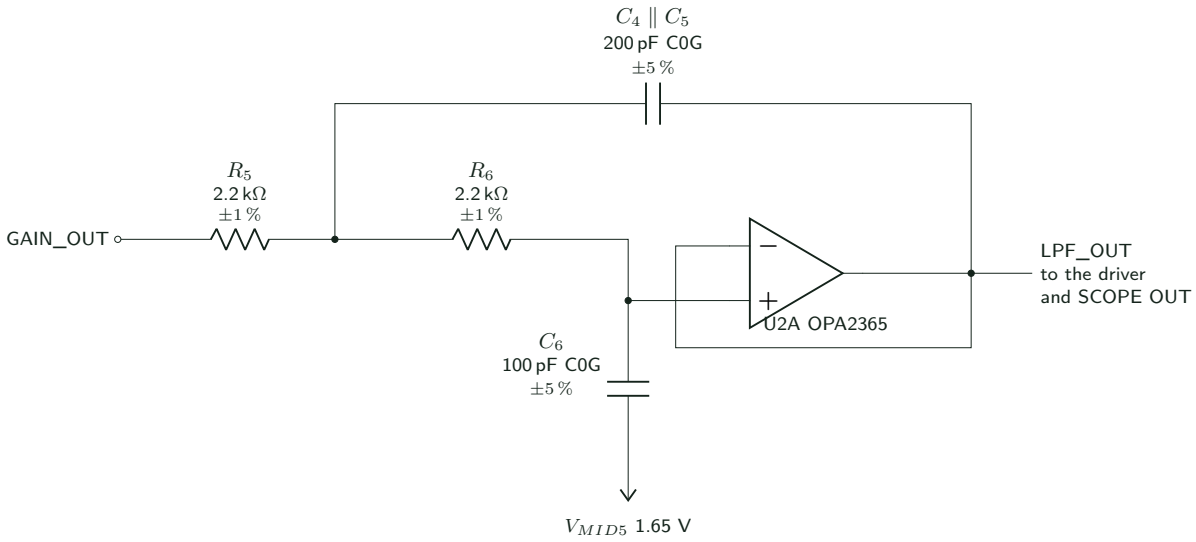
Slide 6: low-pass filter and the whole filter chain

Remove what the ADC cannot represent

ADS7057 test board v3 — study notes

1 The circuit



C4 and C5 (100 pF each) are in parallel, so they act as one $C_1 = 200\,\text{pF}$. C6 is $C_2 = 100\,\text{pF}$. U2A is a follower.

2 Derivation from the node equations

The output equals the + input voltage: $V_o = V_B$ (deviations from VMID5).

Node B, no current into the + input:

$$\frac{V_B - V_A}{R_6} \cdot (-1) + V_B\, sC_6 = 0 \;\Rightarrow\; V_A = V_B\,(1 + sR_6C_6).$$

Node A:

$$\frac{V_A - V_{in}}{R_5} + \frac{V_A - V_B}{R_6} + sC_1\,(V_A - V_o) = 0, \qquad V_o = V_B.$$

Since $V_A - V_B = V_B\, sR_6C_6$, we have $(V_A - V_B)/R_6 = V_B\, sC_6$ and $sC_1(V_A - V_B) = V_B\, s^2R_6C_1C_6$. Then

$$V_A - V_{in} = -R_5V_B\,(sC_6 + s^2R_6C_1C_6) \;\Rightarrow\; V_{in} = V_B\big[1 + sC_6(R_5 + R_6) + s^2R_5R_6C_1C_6\big].$$

$$\boxed{H(s) = \frac{1}{1 + s\,C_6\,(R_5 + R_6) + s^2\,R_5R_6C_1C_6}} \qquad \omega_0 = \frac{1}{\sqrt{R_5R_6C_1C_6}}, \quad Q = \frac{\sqrt{R_5R_6C_1C_6}}{C_6\,(R_5 + R_6)}.$$

With $R_5 = R_6 = 2.2\,\mathrm{k\Omega}$, $C_1 = 200\,\mathrm{pF}$, $C_6 = 100\,\mathrm{pF}$:

$$f_0 = \frac{1}{2\pi \cdot 2.2\,\mathrm{k\Omega}\sqrt{200\,\mathrm{pF} \cdot 100\,\mathrm{pF}}} = 511.5\,\mathrm{kHz}, \qquad Q = \frac{2.2\mathrm{k}\sqrt{2 \times 10^{-20}}}{100\mathrm{p} \cdot 4.4\mathrm{k}} = 0.707.$$

It is the Butterworth low-pass: flat in the band, $-3\,\mathrm{dB}$ at f_0, and $-40\,\mathrm{dB}$ per decade above.

Frequency	250 kHz	500 kHz	800 kHz	2 MHz
Loss of the low-pass alone	0.24 dB	2.8 dB	8.4 dB	23.8 dB

3 Aliasing: why a low-pass before the ADC

A sampler at F_S cannot tell f from $|f - kF_S|$. The sample rate is about 1.2 MSPS (estimate for a 25 MHz SPI clock), so the fold frequency is $F_S/2 = 600\,\mathrm{kHz}$:

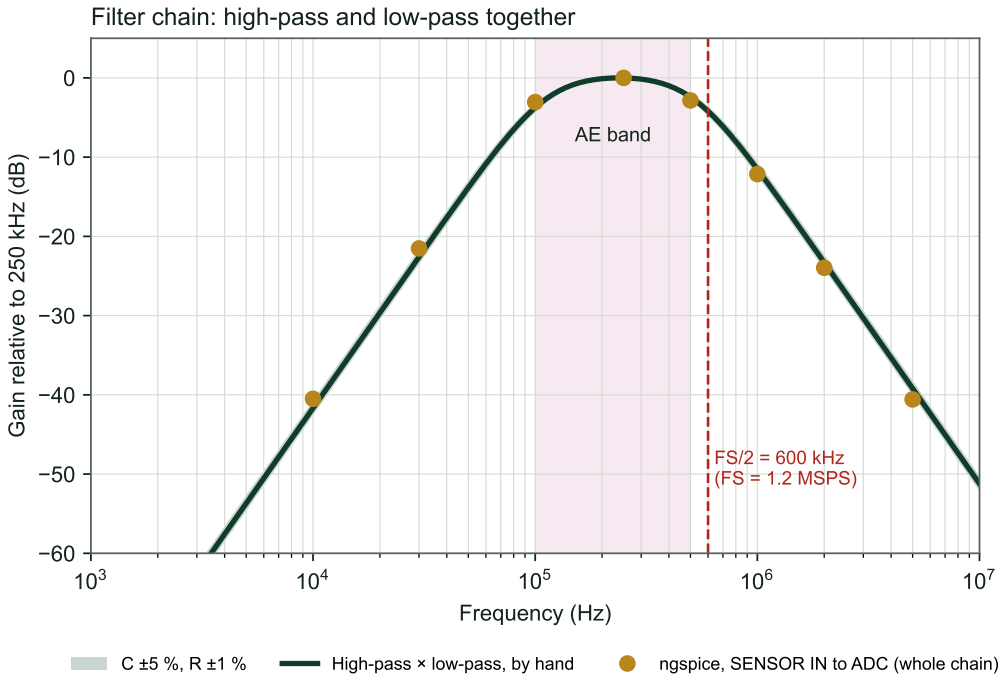
$$f_{\mathrm{alias}} = |f - F_S| : \quad 700\,\mathrm{kHz} \to 500\,\mathrm{kHz}, \quad 800\,\mathrm{kHz} \to 400\,\mathrm{kHz}, \quad 1\,\mathrm{MHz} \to 200\,\mathrm{kHz}.$$

Noise above 600 kHz would land inside the band if the filter did not remove it.

> Be honest about the margin: a second-order filter at 512 kHz removes only 8 dB at 800 kHz and 12 dB at 1 MHz. The design relies on two facts. First, the measured event spectra fall steeply above 300 kHz. Part of that fall came from the perfboard amplifier's own roll-off (its corner was 0.31 MHz), so the real sensor spectrum may extend higher and needs a new measurement. Second, the corner is kept near the top of the band so the pass-band stays flat. A faster SPI clock (up to 2.5 MSPS) moves the fold to 1.25 MHz and adds margin.

4 The whole filter chain

The high-pass (slide 4) and the low-pass multiply, so in dB they add. The result is a band-pass with its $-3\,\mathrm{dB}$ points at 106 kHz and 502 kHz (ngspice, SENSOR IN to ADC). Total gain at 250 kHz is 12.3 dB ($\times 4.1$): the gain stage's 4.9, less the filter and driver losses. Loss at 10 kHz is 40.5 dB and at 2 MHz 24 dB.



The hand model (high-pass times low-pass) and the full ngspice chain agree within 1.1 dB at every point shown.

5 Tolerances

C0G capacitors ±5 % (C14858), resistors ±1 % (C4190). Over all 16 corners:

$$f_0 = 482 \text{ to } 544\,\text{kHz}, \qquad Q = 0.673 \text{ to } 0.743.$$

The upper corner moves about ±6 % and stays above the band edge. Q changes the shape slightly, with at most about 0.4 dB of droop near the corner.